

ePG Introduction to Machine Learning Quiz 1

PCA Explained with Solved Example

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1 Introduction to PCA

Principal Component Analysis (PCA) is a dimensionality reduction technique used to:

- Reduce the number of features
- Remove redundancy
- Retain maximum information (variance)
- Improve visualization and computation speed

PCA transforms correlated variables into a new set of uncorrelated variables called **Principal Components**.

2 Dataset

Consider the following dataset:

$$Z = 2X$$

Hence Z is perfectly correlated with X .

Sample	X	Y	Z
1	1	5	2
2	2	1	4
3	3	4	6
4	4	2	8
5	5	3	10

Thus:

- X and Z are highly correlated
- Y is relatively independent

PCA will try to combine redundant information.

3 Step 1: Mean Calculation

Formula:

$$\mu = \frac{\sum x_i}{n}$$

Mean of X

$$\mu_X = \frac{1 + 2 + 3 + 4 + 5}{5} = 3$$

Mean of Y

$$\mu_Y = \frac{5 + 1 + 4 + 2 + 3}{5} = 3$$

Mean of Z

$$\mu_Z = \frac{2 + 4 + 6 + 8 + 10}{5} = 6$$

4 Step 2: Mean Centering

Subtract feature means from every value.

$$X' = X - \mu_X$$

$$Y' = Y - \mu_Y$$

$$Z' = Z - \mu_Z$$

Centered Dataset

$$A = \begin{bmatrix} -2 & 2 & -4 \\ -1 & -2 & -2 \\ 0 & 1 & 0 \\ 1 & -1 & 2 \\ 2 & 0 & 4 \end{bmatrix}$$

Each column now has mean zero.

5 Step 3: Covariance Matrix

PCA computes covariance matrix:

$$C = \frac{1}{n-1} A^T A$$

where:

- A = centered data matrix

- $n = 5$

Thus:

$$n - 1 = 4$$

Transpose of A

$$A^T = \begin{bmatrix} -2 & -1 & 0 & 1 & 2 \\ 2 & -2 & 1 & -1 & 0 \\ -4 & -2 & 0 & 2 & 4 \end{bmatrix}$$

Matrix Multiplication

$$A^T A = \begin{bmatrix} 10 & -3 & 20 \\ -3 & 10 & -6 \\ 20 & -6 & 40 \end{bmatrix}$$

Now divide by 4:

$$C = \frac{1}{4} \begin{bmatrix} 10 & -3 & 20 \\ -3 & 10 & -6 \\ 20 & -6 & 40 \end{bmatrix}$$

Thus covariance matrix becomes:

$$C = \begin{bmatrix} 2.5 & -0.75 & 5 \\ -0.75 & 2.5 & -1.5 \\ 5 & -1.5 & 10 \end{bmatrix}$$

6 Interpretation of Covariance Matrix

$$C = \begin{bmatrix} 2.5 & -0.75 & 5 \\ -0.75 & 2.5 & -1.5 \\ 5 & -1.5 & 10 \end{bmatrix}$$

- Diagonal entries are variances
- Large covariance:

$$\text{Cov}(X, Z) = 5$$

because:

$$Z = 2X$$

- Negative covariance:

$$\text{Cov}(X, Y) = -0.75$$

indicates inverse variation.

7 Step 4: Eigenvalues and Eigenvectors

PCA computes eigenvalues and eigenvectors of covariance matrix.

PCA solves:

$$Cv = \lambda v$$

where:

- C = covariance matrix
- v = eigenvector
- λ = eigenvalue

Characteristic Equation

$$|C - \lambda I| = 0$$

Thus:

$$\begin{vmatrix} 2.5 - \lambda & -0.75 & 5 \\ -0.75 & 2.5 - \lambda & -1.5 \\ 5 & -1.5 & 10 - \lambda \end{vmatrix} = 0$$

Solving gives approximately:

$$\lambda_1 \approx 12.57$$

$$\lambda_2 \approx 2.43$$

$$\lambda_3 \approx 0$$

8 Step 5: Eigenvectors

Corresponding eigenvectors are approximately:

$$v_1 = \begin{bmatrix} 0.44 \\ -0.13 \\ 0.89 \end{bmatrix}$$

$$v_2 = \begin{bmatrix} 0.27 \\ 0.96 \\ 0.05 \end{bmatrix}$$

$$v_3 = \begin{bmatrix} -0.89 \\ 0 \\ 0.44 \end{bmatrix}$$

Interpretation

- Largest eigenvalue corresponds to PC1
- Second largest eigenvalue corresponds to PC2
- Third eigenvalue is nearly zero because one feature is redundant

Since:

$$Z = 2X$$

the third dimension contains almost no new information.

9 Step 6: Constructing Principal Components

Principal components are linear combinations of original variables.

Principal Component 1

Using eigenvector:

$$v_1 = \begin{bmatrix} 0.44 \\ -0.13 \\ 0.89 \end{bmatrix}$$

we get:

$$PC1 = 0.44X - 0.13Y + 0.89Z$$

Principal Component 2

Using eigenvector:

$$v_2 = \begin{bmatrix} 0.27 \\ 0.96 \\ 0.05 \end{bmatrix}$$

we get:

$$PC2 = 0.27X + 0.96Y + 0.05Z$$

10 Step 7: Variance Explained

Total variance:

$$12.57 + 2.43 + 0 = 15$$

Variance Captured by PC1

$$\frac{12.57}{15} \times 100 = 83.8\%$$

Variance Captured by PC2

$$\frac{2.43}{15} \times 100 = 16.2\%$$

Variance Captured by PC3

$$\frac{0}{15} \times 100 = 0\%$$

Thus:

- PC1 captures most information
- PC2 captures remaining variation
- PC3 contributes almost nothing

Hence PCA keeps:

$$(PC1, PC2)$$

and discards PC3.

11 Step 8: Computing Actual PC Values

Take first centered sample:

$$(-2, 2, -4)$$

Computing PC1

$$PC1 = 0.44(-2) + (-0.13)(2) + 0.89(-4)$$

$$= -0.88 - 0.26 - 3.56$$

$$PC1 = -4.70$$

Computing PC2

$$PC2 = 0.27(-2) + 0.96(2) + 0.05(-4)$$

$$= -0.54 + 1.92 - 0.20$$

$$PC2 = 1.18$$

Thus original point:

$$(1, 5, 2)$$

becomes:

$$(-4.70, 1.18)$$

in PCA space.

12 Step 9: Dimensionality Reduction

Original feature vector:

$$(X, Y, Z)$$

After PCA:

$$(PC1, PC2)$$

Thus:

- 3-dimensional data becomes 2-dimensional
- redundancy is removed
- most information is preserved

13 Final Interpretation

Instead of using original correlated features:

$$(X, Y, Z)$$

machine learning algorithms use:

$$(PC1, PC2)$$

These transformed features:

- are uncorrelated
- capture maximum variance
- reduce dimensionality
- improve efficiency